

SunMint Tree-Growth Monitoring — Video Capture + Python Analysis Worker

Spec v1.2 · 2026-08-25 · Prepared by Sophia Truesight (autopilot) for Gary Teh

Scope: monitor_tree_growth module (DApp + SunMint app) — nearby-tree GPS selection, video capture, Python analysis worker, PM002 carbon math, on-chain event flow.

1. WHY

The SunMint model requires per-tree growth evidence that a VVB can audit. Phone-camera measurement is the proven pattern (TREEO 94–95% DBH accuracy $R^2 \geq 0.95$; ACORN/Agerpoint moving ground-truth to phone; Greenstand photo-verified). Video adds stronger tamper-evidence than a single photo — the walk-around cadence (farm-build game) is captured continuously.

2. SYSTEM ARCHITECTURE

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Farmer phone (SunMint app + main DApp monitor page) Backend (Python worker)

- open monitor module → GPS read ■■ [ingest] receive video + metadata
- rank nearby trees closest → furthest (Haversine) ■■ [hash] SHA-256 of video → audit trail
- pick tree from dropdown ■■ [process] OpenCV:
- record video: walk-around, calibration card visible ■ frame extraction (e.g. 1 fps)
- auto-extract GPS + timestamp + farmer RSA signature ■ calibration-card detection (ARUCO/contour)
- save draft locally (offline-first) ■ DBH estimation (pixel ratio → trig)
- POST signed metadata + video → /api/monitor/ingest ■ species lookup → allometric formula
- → AGB → BGB ($R=0.32$) → $\times 0.47$ → $\times 44/12$
- [emit] signed [TREE GROWTH MONITORING EVENT] → Edgar
- [notify] farmer growth result in app

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Key decisions

- Capture + lightweight math in browser; heavy CV in Python. GAS is glue only (relay/notify), NOT the analysis engine (no OpenCV runtime).
- Hash the video before processing → reproducible, tamper-evident pipeline (VVB-grade).
- Offline-first — farmers in the field record, queue, upload when online (SunMint mobile is already offline-first).
- **Dual deployment:** the monitor page ships to BOTH the main DApp (dapp.truesight.me) AND the SunMint app (sunmint.truesight.me).

3. MODULE A — monitor_tree_growth (DApp + SunMint app)

3.1 Page: monitor_tree_growth.html — deployed to BOTH DApp targets

- **Main DApp:** dapp_beta/monitor_tree_growth.html → dapp.truesight.me/monitor_tree_growth.html
- **SunMint app:** truesight_me_beta/sunmint/monitor-tree-growth/index.html → sunmint.truesight.me/monitor-tree-growth/
- Same codebase/behavior in both; crypto + offline queue identical (WebCrypto RSA-2048 keypair in localStorage, IndexedDB draft queue)
- **Nearby tree selection:** GPS-based ranking closest → furthest (Haversine) → dropdown → pick tree → Start Monitoring Video
- Select farm → select/create tree record (per-tree ID, species, planting date)
- Video capture: `<input type="file" accept="video/*" capture="environment">` + live preview via getUserMedia (existing pattern in report_tree_planting.html)
- Recording guidance overlay: hold card against tree at 1.3 m (breast height), portrait, walk 360° slowly
- Auto-metadata: GPS, timestamp, farmer public key
- Offline queue: IndexedDB draft storage; upload on reconnect
- Signed POST: farmer RSA signature over metadata + video hash

3.1.1 Nearby Tree Selection (UX flow)

- On opening the monitor module, the app reads the device GPS (existing geolocation permission)
- Pulls the farmer's registered trees (from farm/tree registry; each tree has lat/lng from the planting event)
- **Ranks nearby trees closest → furthest** (Haversine distance from current GPS)
- Farmer picks a tree from the dropdown → tree record selected → **Start Monitoring Video** button enables
- Fallbacks: manual search by tree ID; GPS unavailable → "All trees" list (alphabetical / last-measured first); offline → last-known GPS + cached registry
- Distance badge per tree (e.g. "12 m", "48 m", "203 m") so the farmer can walk to the nearest tree

3.1.2 GPS proximity sort (Haversine)

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$$\text{dist} = 2R \cdot \text{asin}(\sqrt{\sin^2(\Delta\text{lat}/2) + \cos(\text{lat1}) \cdot \cos(\text{lat2}) \cdot \sin^2(\Delta\text{lng}/2)})$$
 R = 6371 km

...

- Client-side JS; pure function, unit-testable; no backend round-trip needed

3.2 Data contract (request)

```json

{

"farmer\_id": "<did/farm>",

```
"tree_id": "SUNMINT-0001",
"species": "cacao | brazil_nut | acai | mahogany | jatoba",
"video_b64": "<base64 or multipart>",
"video_sha256": "<hex>",
"gps": {"lat": -14.5, "lng": -39.0},
"captured_at": "2026-08-25T14:30:00Z",
"farmer_signature": "<rsa-sig over sha256 of metadata+video_hash>",
"nearby_trees": [{"tree_id": "SUNMINT-0001", "distance_m": 12}] // context: sorted closest→furthest at capture
}
...
```

### 3.3 Data contract (response)

```
...
{ status, measurement_id, dbh_cm, height_m, agb_kg, co2e_kg,
 confidence, analysis_sha256, edgar_event }
...
```

## 4. MODULE B — Python analysis worker (truesight\_autopilot)

### 4.1 Stack

- Python 3.11+, OpenCV (frame extraction, ARUCO card detection), numpy, hashlib
- Optional: scikit-image / lightweight YOLO for tree segmentation (v2 polish)
- Crypto: farmer public key verification (RSA-2048), result signing (Sophia DAO key)
- Deployment: on the autopilot box (or a small EC2 worker) exposed as /api/monitor/ingest

### 4.2 Pipeline (pure functions, unit-testable)

1. `extract_frames(video, fps=1) → frames[]`
2. `detect_calibration_card(frame) → card corners (ARUCO marker or contour; ISO 7810 85.6×53.98 mm)`
3. `estimate_dbh(frame, card_bounds, px_to_mm) → DBH at 1.3 m (similar-triangle / pixel-ratio)`
4. `allometric_agb(species, dbh) → AGB kg (species table; IPCC/Chave allometrics)`
5. `carbon_math(agb) → PM002 chain:  $\Delta\text{BGB} = \Delta\text{AGB} \times R$  ( $R=0.32$ ) ·  $\text{CO}_2\text{e} = ((\Delta\text{AGB} + \Delta\text{BGB}) \times 0.47) \times 44/12 \times (1 - A_{\text{pre}})(1 - A_{\text{unc}})(1 - \text{LD})(1 - \text{AR})(1 - \text{RB}) - (E_{\text{proj}} - E_{\text{base}}) \cdot \text{AR}=10\% \cdot \text{RB}=20\%$`
6. `sign_result(measurement) → measurement_id + signature`
7. `emit_event(result) → [TREE GROWTH MONITORING EVENT] → Edgar`

### 4.3 Allometric species table (v1 seeds; needs field calibration)

| Species                 | Formula source                                | Notes            |
|-------------------------|-----------------------------------------------|------------------|
| Cacao (Theobroma cacao) | Chave et al. tropical / local Bahia equations | needs validation |
| Brazil nut              | Chave tropical moist                          | —                |
| Acai                    | palm-specific AGB                             | —                |
| Mahogany                | Chave tropical                                | —                |
| Jatoba                  | Chave tropical                                | —                |

## 5. EVENT TYPES

- [TREE GROWTH MONITORING EVENT] — new event; per-measurement on-chain record: tree\_id, dbh, agb, co2e, gps, captured\_at, analysis\_sha256, farmer\_sig, worker\_sig
- Reuses existing [TREE PLANTING EVENT] for the initial planting record
- Plan Vivo / VVB audit path: raw video (stored) + hash + analysis log = full provenance

## 6. IMPLEMENTATION ROADMAP (phased)

| Phase                  | Deliverable                                                                                                                                                                                         | Effort | Owner            | Exit criteria                              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------|--------------------------------------------|
| P0 — Spec & contract   | This spec; API contract frozen                                                                                                                                                                      | 0.5 d  | Sophia           | Gary sign-off                              |
| P1 — Browser capture   | monitor_tree_growth.html in dapp_beta<br>AND sunmint/monitor-tree-growth in truesight_me_beta:<br><b>nearby-tree GPS ranking (Haversine) + dropdown</b> , video capture, offline queue, signed POST | 4–5 d  | Dev (beta-first) | UAT on both beta sites                     |
| P2 — Python worker MVP | ingest → frames → card detect → DBH → CO <sub>2</sub> e → signed result                                                                                                                             | 3–5 d  | Sophia/Dev       | Unit tests pass (compileall, ruff, pytest) |
| P3 — Integration       | worker endpoint ↔ dapp POST; response contract                                                                                                                                                      | 2 d    | Dev              | End-to-end on beta                         |
| P4 — On-chain event    | [TREE GROWTH MONITORING EVENT] in dao_protocol/Edgar                                                                                                                                                | 2 d    | Dev              | Event lands on ledger                      |

| Phase                    | Deliverable                                         | Effort  | Owner      | Exit criteria           |
|--------------------------|-----------------------------------------------------|---------|------------|-------------------------|
| P5 — Growth history view | per-tree measurements over time; farmer dashboard   | 2–3 d   | Dev        | History render on beta  |
| P6 — Field calibration   | species allometrics vs tape/calipers on pilot plots | ongoing | Field team | $R^2 \geq 0.9$          |
| P7 — VVB audit prep      | analysis reproducibility doc + video retention      | 1–2 d   | Sophia     | VVB-ready evidence pack |

## 7. CHECKLIST (execution tracking)

- ☐ P0: spec signed off by Gary
- ☐ P1: monitor\_tree\_growth.html drafted (dapp\_beta PR)
- ☐ P1: sunmint/monitor-tree-growth/index.html drafted (truesight\_me\_beta PR) — sunmint.truesight.me target
- ☐ P1: nearby-tree GPS ranking (Haversine) + dropdown implemented
- ☐ P1: offline queue (IndexedDB) implemented
- ☐ P1: UAT on beta sites
- ☐ P2: worker scaffold in truesight\_autopilot
- ☐ P2: extract\_frames + detect\_calibration\_card
- ☐ P2: estimate\_dbh + allometric\_agb + carbon\_math (PM002)
- ☐ P2: sign\_result + emit\_event
- ☐ P2: LOCAL TEST SUITE PASSES (compileall, ruff, ruff format, pytest)
- ☐ P3: endpoint wired to DApp POST
- ☐ P4: Edgar event type registered
- ☐ P5: growth history view on beta
- ☐ P6: field calibration data collected
- ☐ P7: VVB evidence pack assembled

## 8. RISKS & OPEN QUESTIONS

- Card detection in harsh light / occlusion → ARUCO markers more robust than plain contours; fallback to manual DBH entry + photo evidence (TREEO-style)
- Species allometrics for cacao are weak → P6 field calibration is critical; Plan Vivo allows project-specific equations
- Video size (50–200 MB) → compress to  $\leq 720p$  / 10 fps in app before upload; worker cap 5 min video
- VVB acceptance of phone-video MRV → Plan Vivo participatory monitoring precedent (CommuniTree since 2010); keep raw video + hashes

- GPS accuracy ( $\pm 5$ – $10$  m) may mis-rank trees on dense plots → distance badges help; nearest-3 shown with equal priority
- Worker deployment → autopilot box first; scale to dedicated worker if volume demands